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Unprecedented combination of specific tensile strength and ductility achieved in a novel duplex low-density steel

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ABSTRACT

By annealing a cold rolled 30Mn-13Al-1.3C-5Cr steel at 1000 °C, we successfully developed a novel lightweight steel having the tensile strength of 1250 MPa and total elongation of 41.4 % at the measured ultralow density of 6.36 g·cm^{-3}. Such an unprecedented strength-ductility-density synergy results from the new designed austenite-ferrite duplex microstructures, where the granular α ferrite grains can restrain the austenite grain growth, δ ferrite bands capture and deflect the propagating cracks, and high contents of Mn/Al/C in recrystallized austenite result in the extensive precipitation of κ-carbide nanoparticles for strengthening.

Due to the high demand for weight saving and energy conservation, the lightweight Fe-Mn-Al-C steels have received an increasing attention in many industries [1–3] due to their excellent combination of specific strength and ductility [2,4,5]. Among all the relevant researches, Moon et al. [6] set a record on their developed 29.05Mn-12.01Al-1.48C-5.13Cr (weight percentage unless mentioned elsewhere) austenitic steel, which has the specific ultimate tensile strength (SUTS) of 169.9 MPa·g^{-1}·cm^3 and total elongation (TE) of 43.8 % at the density of 6.34 g·cm^{-3}. To our knowledge, this is almost the lowest density achieved in a highly deformable Fe-Mn-Al-C austenitic steel with the SUTS up to 170 MPa·g^{-1}·cm^3. A further improvement on specific strength via lowering density is clearly more challenging, because lower density requires higher contents of Al/C added, which often lead to the brittle intergranular κ-carbides formed at grain boundaries, severely deteriorating ductility [7,8]. A promising pathway of further improving the specific strength-ductility synergy at such a low density is in lack but strongly desired.

Ferrite is more effective on reducing density than austenite because body centered cubic (BCC) structure has lower atomic density than face centered cubic (FCC) [1,9]. Meanwhile, the presence of ferrite phase results in an inhibition on the growth of austenite grains and many austenite/ferrite interfaces, which may improve yield strength (YS) and the strain hardening capacity during deformation [10]. Therefore, it is expected that the austenite-ferrite duplex structure may have greater potential of improving the strength-ductility-density synergy than the mere austenitic one. However, the reported austenite-ferrite duplex steels all have the density higher than 6.50 g·cm^{-3} [11–20], to our knowledge, and it has been instinctively believed that cracks may be easily initiated at the austenite/ferrite interfaces and then propagate along them, thus dramatically deteriorating ductility [21]. For an example, Sohn et al. [13] reported that the increase of ferrite fraction from 13.2 % to 32.3 % in 12Mn-5.5Al-0.7C duplex steel (with the density of 7.20 g·cm^{-3}) led to the increases of the SUTS from 139.3 to 196.5 MPa·g^{-1}·cm^3 but the significant decrease of TE from 45.9 % to 8.2 %. This, however, may also suggest that it shall be possible to improve the SUTS-TE combination by optimizing the presence of ferrite. Therefore, we consider to redesign the previous ultralow density steel (29.05Mn-12.01Al-1.48C-5.13Cr [6]) for the austenite-ferrite duplex microstructure, through which SUTS can be further enhanced while low density and large ductility both remain.

The chemical composition of redesigned austenite-ferrite duplex steel is 30.35Mn-12.60Al-1.32C-4.83Cr, where the addition of about 5 % Cr is to suppress the formation of the intergranular κ-carbides [22,23]. This steel has an ultralow density of 6.36 g·cm^{-3}, close to the previous record value of 6.34 g·cm^{-3} [6], and measured by buoyancy balance according to Archimedes Principle. It was melted in a vacuum induction furnace under argon atmosphere and cast into ingot, which was forged to thick slab and further hot rolled (HR) to 6 mm thickness after the homogenization at 1200 °C for 2 h. The HR plates were heated to 1050 °C and isothermally held for 0.5 h, quenched in water and then cold

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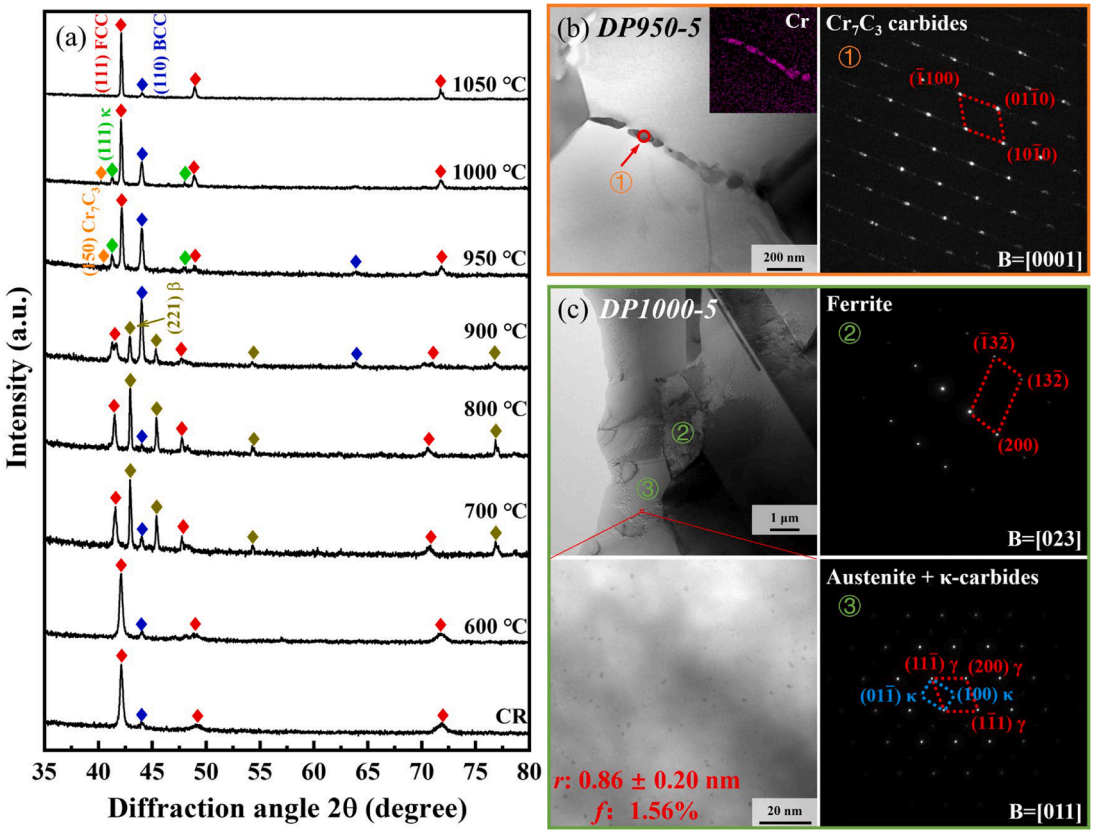


Fig. 1. Microstructures formed in the developed duplex steel after the annealing of the CR structure at 600–1050 °C for 5 min, examined by XRD (a) and TEM (b, c). Cr_7C_3 carbides were identified in DP950–5 (b), and the intragranular κ-carbide nanoparticles were identified in DP1000–5 (c).

rolled (CR) to 3 mm thickness. Next, the CR sheets were annealed at the temperatures ranged from 600 °C to 1050 °C for the various periods in the range of 1–5 min and then quenched in water again. The resultant specimens are termed DP*XXX*–*Y* for easy discussion, where *XXX* and *Y* represent the annealing temperature and period respectively.

Flat dog-bone-shaped tensile specimens with a gage length of 25 mm were fabricated with the length direction along the rolling direction (RD) except for one along the transverse direction (TD) for comparison. Uniaxial tensile tests were conducted at a strain rate of 10^{-3} s^{-1} at room temperature. Three tensile measurements were made for each heat treatment condition and the average values taken. Nanoindentation tests were conducted on the etched sample for studying the mechanical response of different microstructural constituents. The microstructures were examined by X-ray diffraction (XRD), scanning electron microscope (SEM, ZEISS Gemini 300), transmission electron microscope (TEM, FEI Tecnai G20), and electron backscatter diffraction (EBSD, Scanning Auger Nanoprobe, PHI 710). The ferrite fraction was determined from the XRD results, as detailed in supplementary materials.

Fig. 1**a** shows the phase constituents formed in the designed duplex steel after the annealing at 600–1050 °C for 5 min, as measured by XRD. The diffraction peaks of austenite (FCC) and ferrite (BCC) were clearly observed in the CR sample. Although 600 °C is a typical ageing temperature for the precipitation of κ-carbides in Fe-Mn-Al-C steels [24–26], the latter was not detected due to their limited fraction. β-Mn phase was found at 700–900 °C, which is consistent with the previous results as well [25,27,28]. It is a hard phase usually formed at grain boundaries, leading to the brittle fracture [29]. The annealing at 900 °C for 5 min led to clearly strongest BCC diffraction peaks among all the annealing processes, suggesting that austenite-to-ferrite transformation should occur during the heating to this temperature. κ-carbides and Cr_7C_3 carbides both appeared at 950–1000 °C but disappeared at 1050 °C. It should be noted that the formation of Mn/Al/C-rich κ-carbides above 950 °C consumes a large quantity of Mn/Al/C atoms, leading to the reduced lattice parameters of austenite and then the shift of its diffraction peaks towards higher 2θ, as have been reported in Refs. [30,31]. The calculated equilibrium phase fractions at the temperature range of 550–1400 °C are shown in **Fig. S1**, where the measured ferrite fractions are also included for comparison. The calculated ferrite fraction remains almost constant about 33 % between 991 °C and 1250 °C and then increases to the maximum when temperature decreases from 991 °C to 854 °C. In contrast, the largest ferrite fraction measured by XRD appears at 900 °C. However, the measured ferrite fraction actually keeps decreasing when temperature increases above 900 °C, suggesting that ferrite-to-austenite transformation should occur.

XRD results in Fig. 1**a** indicate that both Cr_7C_3 carbides and κ-carbides should be formed during the annealing at 950–1000 °C for 5 min. We observed two types of Cr_7C_3 carbide particles: one has the coarse size of several micrometers and is located in ferrite bands (see **Fig. S2a**), the other has the size of hundreds of nanometers and precipitates at grain boundaries, see Fig. 1**b**. κ-carbides can be formed at grain boundaries and the BCC/FCC interfaces with the relatively coarse size of 2–3 μm in DP950–5, see **Fig. S2a**, or within austenite grains with the ultrafine size of 0.86 ± 0.20 nm in DP1000–5 (see Fig. 1**c**). The former usually leads to a brittle fracture whilst the latter formed during water quenching contributes to precipitation strengthening without significantly sacrificing plasticity [1]. When the annealing temperature increases to 1000 °C, both Cr_7C_3 carbides and the intergranular κ-carbides are significantly dissolved (see Fig. 1**c** and **Fig. S2b**).

Fig. 2 shows the microstructural evolution from cold rolling to the annealing at 950–1050 °C for 1–5 min. It can be clearly seen that the ferrite grains in the CR specimen also have two different morphologies and sizes. One is granular and often located at the triple points of austenite grain boundaries with the size of 0.6–1.8 μm; the other is band-like and elongated along the RD with the length of 12–30 μm and the

Table 1
Tensile properties of the resultant specimens with the microstructural constituent fractions. (S)YS, (S)UTS, and TE represent (specific) yield strength, (specific) ultimate tensile strength, and total elongation, respectively. Rex-γ, α, and δ are recrystallized austenite, α ferrite, and δ ferrite, respectively.

Properties	CR	DP950–1	DP950–5	DP1000–1	DP1000–5	DP1000–5-TD	DP1050–1
YS (MPa)	–	1015 ± 5	920 ± 15	1075 ± 10	1035 ± 5	1005 ± 15	905 ± 15
UTS (MPa)	–	1440 ± 10	1350 ± 15	1295 ± 15	1250 ± 15	1190 ± 15	1205 ± 10
TE (%)	–	5.2 ± 0.2	5.5 ± 0.2	27.4 ± 0.7	41.4 ± 0.5	18.1 ± 0.4	47.5 ± 1.0
SYS (MPa·g^{-1}·cm^3)	–	159.6 ± 0.8	144.7 ± 2.4	169.0 ± 1.6	162.7 ± 0.8	158.0 ± 2.4	142.3 ± 2.4
SUTS (MPa·g^{-1}·cm^3)	–	226.4 ± 1.6	212.3 ± 2.4	203.6 ± 2.4	196.5 ± 2.4	187.1 ± 2.4	189.5 ± 1.6
Rex-γ fraction (%)	7.9	19.4	54.8	91.6	95.2	95.2	96.8
γ grain size (μm)	15.38 ± 16.47	14.64 ± 8.88	6.26 ± 4.50	5.13 ± 4.74	7.40 ± 5.13	7.40 ± 5.13	9.65 ± 8.17
Ferrite fraction by XRD (%)	12.6	–	30.5		17.8	17.8	–
α fraction by EBSD (%)	5.7	19.6	23.8	9.1	4.1	4.1	2.7
δ fraction by EBSD (%)	5.4	9.8	12.1	10.5	9.3	9.3	4.6

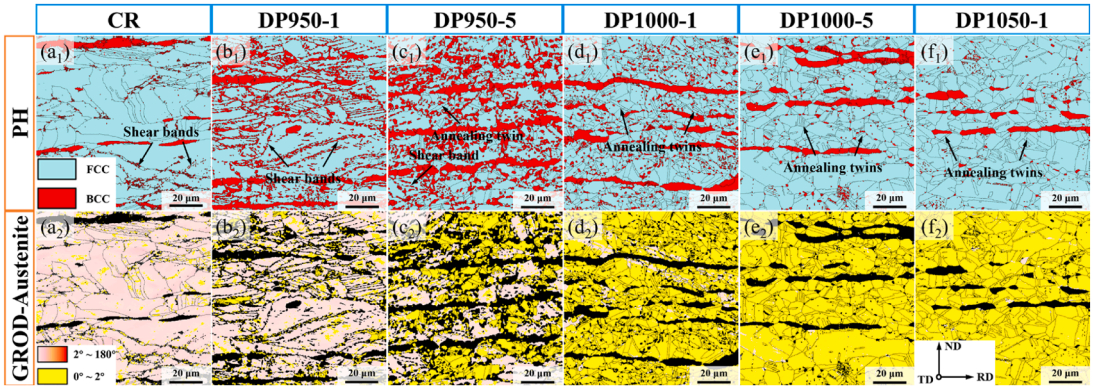


Fig. 2. EBSD results on the microstructures formed in the developed duplex steel after the different annealing processes. (a_1-f_1) phase (PH) mappings and (a_2-f_2) grain-reference-orientation-deviation (GROD) mappings of austenite, where dark lines in austenite are high-angle grain boundaries with the misorientation more than 15°. TD, RD, and ND represent the transverse, rolling, and normal direction, respectively.

thickness of 2–6 μm. The latter can be traced to the preceding HR structure and is still present after the solution treatment at 1200 °C for 1 h (see **Fig. S3**), indicating that they should be δ ferrite that was initially formed during solidification, since this has been found in the previous researches on Fe-Mn-Al-C steels [11,17]. The measured fractions of recrystallized (Rexed) austenite, granular α ferrite, and banded δ ferrite in all the specimens are summarized in Table 1, where the Rexed fraction in austenite was determined from the grain-reference-orientation-deviation (GROD) mappings by setting the GROD value less than 2° [32,33].

The annealing at 950 °C for 5 min leads to partial recrystallization in austenite with the Rexed fraction of 54.8 % and the ferrite fraction up to 36 % (as determined by EBSD in Fig. 2, *i.e.*, the corresponding δ ferrite bands are thickest, and the granular α ferrite grains are coarsest in DP950–5. This suggests austenite-to-ferrite transformation may occur during the heating to 950 °C for 5 min, and ferrite-to-austenite transformation may occur above this temperature, both are consistent with the XRD results in Fig. 1**a**. Meanwhile, Cr_7C_3 carbides and the intergranular κ-carbides are also formed during the annealing at 950 °C (see Fig. 1**b** and **Fig. S2a**), *i.e.*, austenite → ferrite + Cr_7C_3 carbides + the

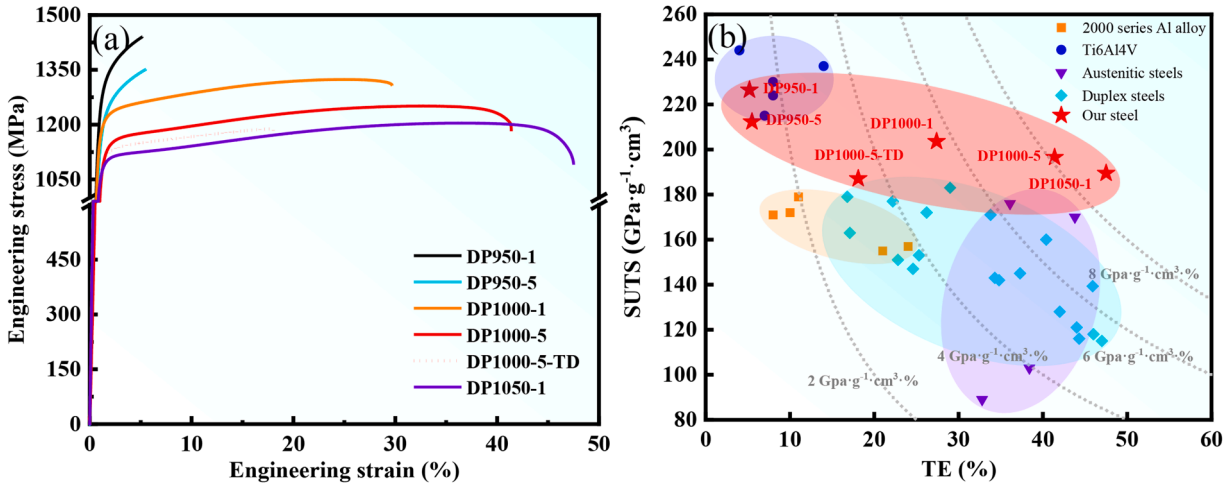


Fig. 3. Engineering stress-strain curves of our designed duplex lightweight steel subjected to the different annealing processes (a), and comparison of tensile properties of our developed lightweight steel with other existing high-strength metallic materials (b), where the contour lines indicate the product of SUTS and TE ranged from 2 to 8 GPa·g^{-1}·cm^3. SUTS and TE represent specific ultimate tensile strength and total elongation, respectively.

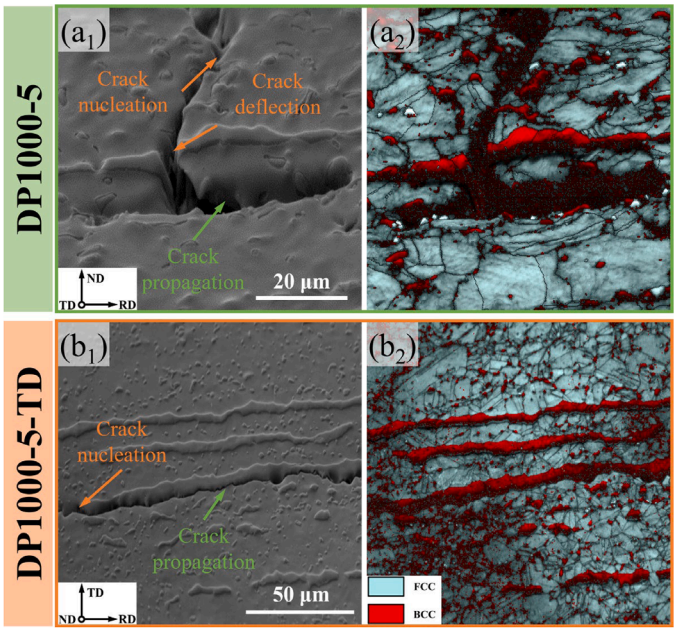


Fig. 4. SEM and EBSD images on the typical cracks in DP1000–5 (a) and DP1000–5-TD (b). FCC and BCC are austenite and ferrite, respectively. TD, RD, and ND represent the transverse, rolling, and normal direction, respectively.

intergranular κ-carbides. Nevertheless, the increase of the annealing temperature from 950 °C to 1050 °C leads to thinner δ ferrite bands and finer granular α grains with lower fraction of ferrite (see Fig. 2 and Table 1). Meanwhile, Cr_7C_3 carbides and the intergranular κ-carbides start to dissolve significantly at 1000–1050 °C.

The annealing at 1000 °C or above also results in a complete recrystallization and the subsequent grain growth, as seen in Fig. 2**d-f** and Table 1**.** The Rexed austenite grain size increases from 5.13 ± 4.74 μm in DP1000–1 to 7.40 ± 5.13 μm in DP1000–5 and 9.65 ± 8.17 μm in DP1050–1, because the number and fraction of granular α ferrite both decrease with the increases in the annealing temperature and period, and the presence of the granular α ferrite grains at the boundaries and triple points can effectively retard the austenite grain growth. Recrystallization wipes off the deformed structure such as shear bands and grain growth should significantly reduce the quantity of (sub)grain boundaries so that the nucleation sites for the granular α ferrite grains decrease prominently; consequently, the number and fraction of granular α both decrease more than those of δ bands. In summary, the annealing at 950 °C produces the coarsest δ and α grains (Fig. 2**c**1), which then partially transform back to austenite when the annealing temperature is further raised.

Fig. 3**a** shows the engineering stress-strain curves of the annealed specimens. Their specific strength and ductility are also summarized in Table 1. It is seen that DP950–1 has the highest ultimate tensile strength (UTS) but lowest TE. The increase of the annealing time to 5 min led to the decrease in both YS and UTS by about 90 MPa but no improvement in ductility. In contrast, the increase of the annealing temperature to 1000 °C results in YS and TE both greatly improved. The nanoindentation results (see **Fig. S4**) on DP1000–5 show that ferrite is clearly softer than the Rexed austenite, leading to lower YS measured in the specimens annealed at 950 °C than those annealed at higher temperatures. Moreover, the presence of such a large amount of ferrite in DP950–1 and DP950–5 causes the incompatible straining at many interfaces between ferrite and the unRexed austenite due to the great difference in their strength/hardness, leading to the easy initiation and propagation of cracks at/along the interfaces; therefore, both specimens have poor ductility. When the annealing temperature increases from 950 °C to 1000 °C, the CR austenite is almost fully Rexed and the total ferrite fraction decreases significantly so that the difference in the strength of FCC and BCC and the quantity of FCC/BCC interfaces both decrease; thus, the harder austenite as the major phase contributes more to YS and both FCC/BCC phases can deform more compatibly when compared to the specimens annealed at 950 °C so that both YS and TE are greatly improved. However, the further increase of the annealing temperature to 1050 °C causes TE improved but YS/UTS remarkably impaired, which are primarily because of the austenite grains growth (see Table 1) and the significant dissolution of Cr_7C_3 carbides/the intergranular κ-carbides, as revealed in Fig. 1**a**.

Both DP1000–5 and DP1050–1 specimens exhibit clearly superior mechanical combination to the existing high-strength metallic materials, see Fig. 3**b**, because their products of SUTS and TE are more than 8 GPa·g^{-1}·cm^3·%; whilst less than 4 GPa·g^{-1}·cm^3·% for 2000 series aerospace aluminum alloys [34] and titanium alloy of Ti6Al4V [35], and in the ranges of 2.7–6.5 and 2.9–7.4 GPa·g^{-1}·cm^3·% for the duplex [12,13,16,19,20] and austenitic Fe-Mn-Al-C steels [15,36], respectively. In other words, the designed duplex steel has higher specific tensile strength than the Al alloys and other low-density steels at the similar level of ductility, and much more ductile than Ti6Al4V alloy whilst its SUTS is just slightly lower.

The tensile properties of DP1000–5 were also examined along the TD (the resultant specimen termed DP1000–5-TD), and the results are given in Fig. 3**a** and Table 1. It has the similar YS/UTS but lower TE when compared to DP1000–5. Since only δ ferrite bands have the clear orientation among all the microstructural constituents, they are most likely the reason for lower ductility measured along the TD. In Fig. 4**a**, a tensile crack in DP1000–5 was supposed to be nucleated at the α/γ interface and then propagate along the normal direction (ND) until it encountered a δ ferrite band parallel to the RD, where it was deflected to propagate along the RD. These RD cracks may propagate into the soft δ ferrite bands and then get blunt through dislocation emission and local plastic dissipation in vicinity of the crack tip [37], which significantly delocalized the plastic strain [38]. Therefore, the presence of more δ ferrite bands increases the frequency of capturing and deflecting the ND cracks, the latter are the main cause of tensile failure since thickness is the smallest dimension according to the gauge geometry of tensile specimens. In DP1000–5-TD, however, the δ/γ interfaces are perpendicular to the tensile direction (see Fig. 4**b**) and subjected to the high de-cohesive stress; consequently, cracks were initiated at these δ/γ interfaces in addition to the α/γ interfaces, and then propagated until they quickly got connected across the full thickness during straining, leading to poor ductility.

In summary, we successfully developed a new Fe-Mn-Al-C-Cr austenite-ferrite duplex steel having both the ultralow density of 6.36 g·cm^{-3} and the ultrahigh product of SUTS and TE exceeding 8 GPa·g^{-1}·cm^3·%. Such an unprecedented mechanical combination was achieved by the duplex microstructures comprising two types of ferrites, *i.e.*, the granular α ferrite grains plus δ ferrite bands, and fully Rexed austenitic matrix, which was realized by the annealing of the CR structure at 1000 °C for 5 min. The granular α ferrite grains can restrain the austenite grain growth when they are sufficient though cracks are also easily initiated at the α/γ interfaces during deformation; however, these cracks can be captured and deflected by δ ferrite bands, where they even get blunted. Meanwhile, Cr_7C_3 carbides, the intergranular κ-carbides, and ferrite significantly transformed to austenite during the annealing at 1000 °C, resulting in high Mn/Al/C concentrations in austenite for the high-density nanosized κ-carbide precipitation during water quenching. Therefore, our work demonstrates that the duplex microstructural design is a promising method of achieving the excellent strength-ductility-density synergy for low-density steels.

CRediT authorship contribution statement

Zhipeng Liu: Writing – original draft, Methodology, Investigation. **Cancan Ding:** Methodology. **Ru Ge:** Investigation. **Bin Hu:**

Methodology. **Haiwen Luo:** Writing – review & editing, Resources, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.scriptamat.2025.116668.

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